Computer Vision COMP90086

Workshop Week 6

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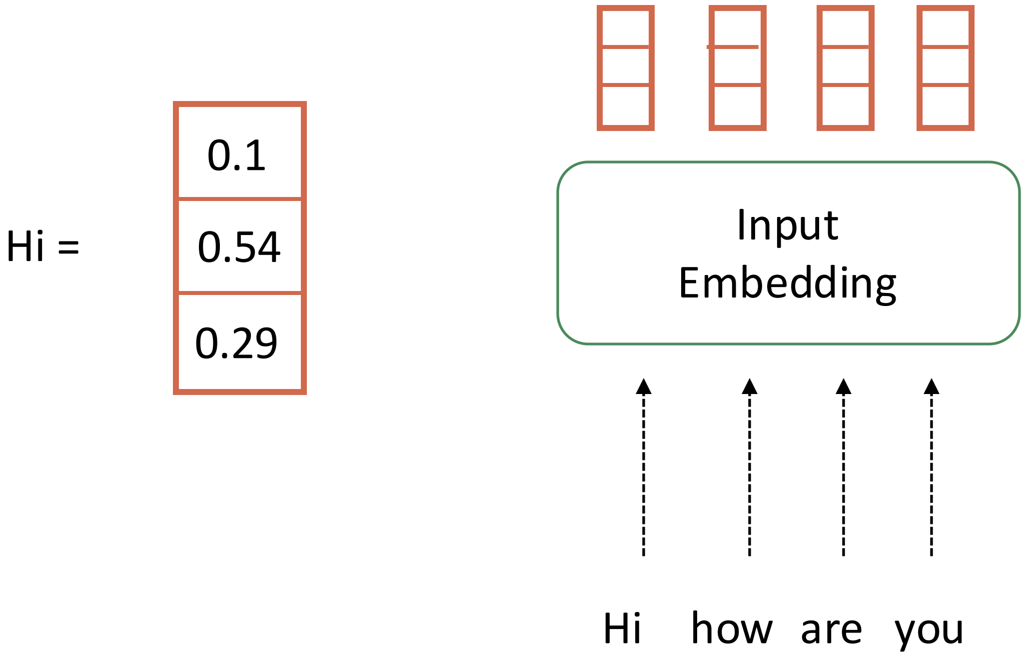
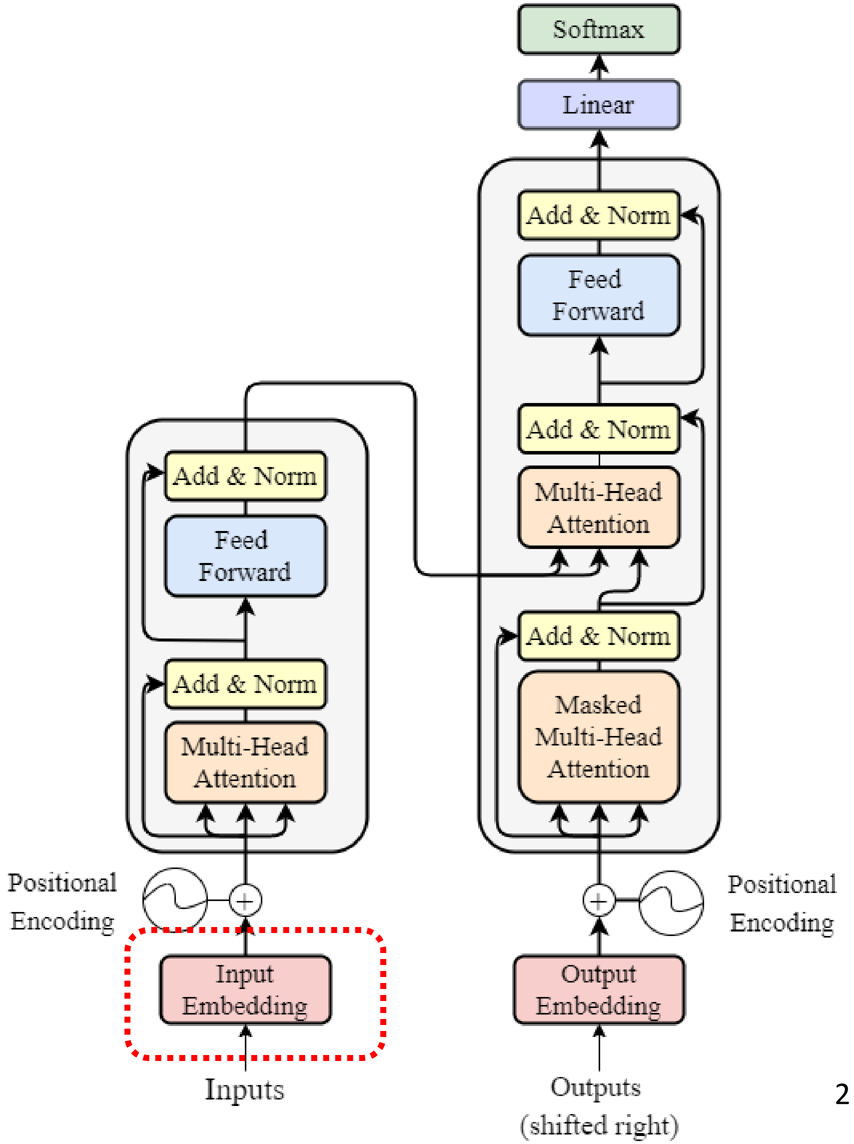
thixuanmay.le@unimelb.edu.au

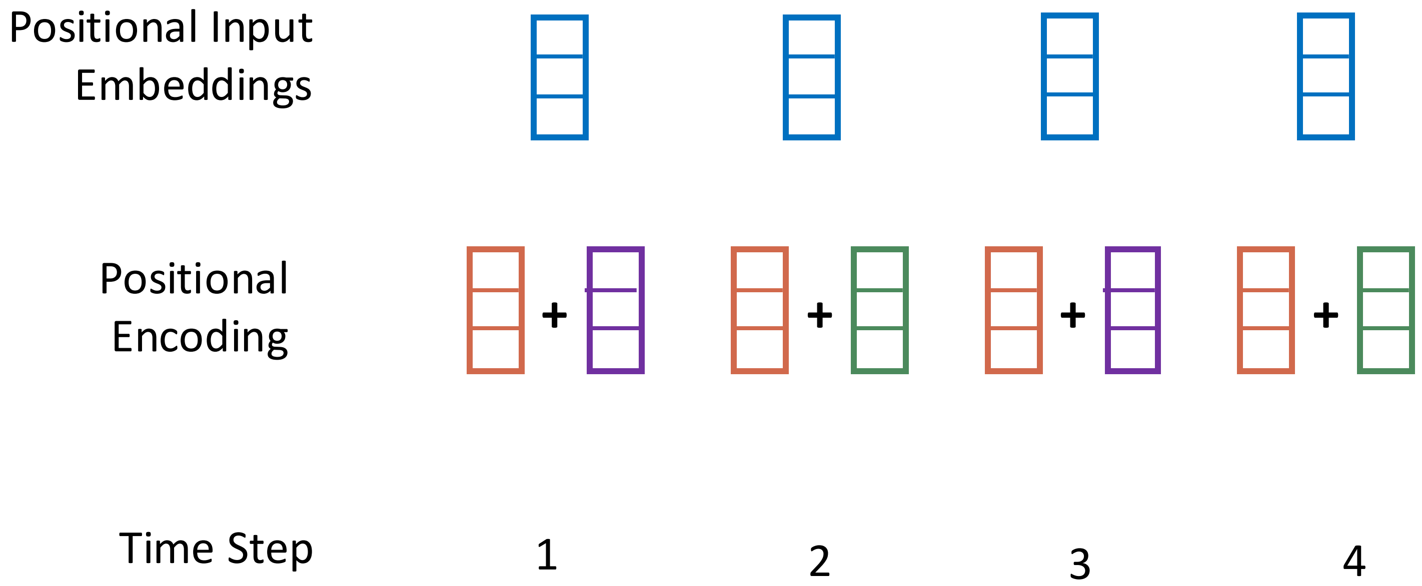
Transformer Model

1. Input Embedding



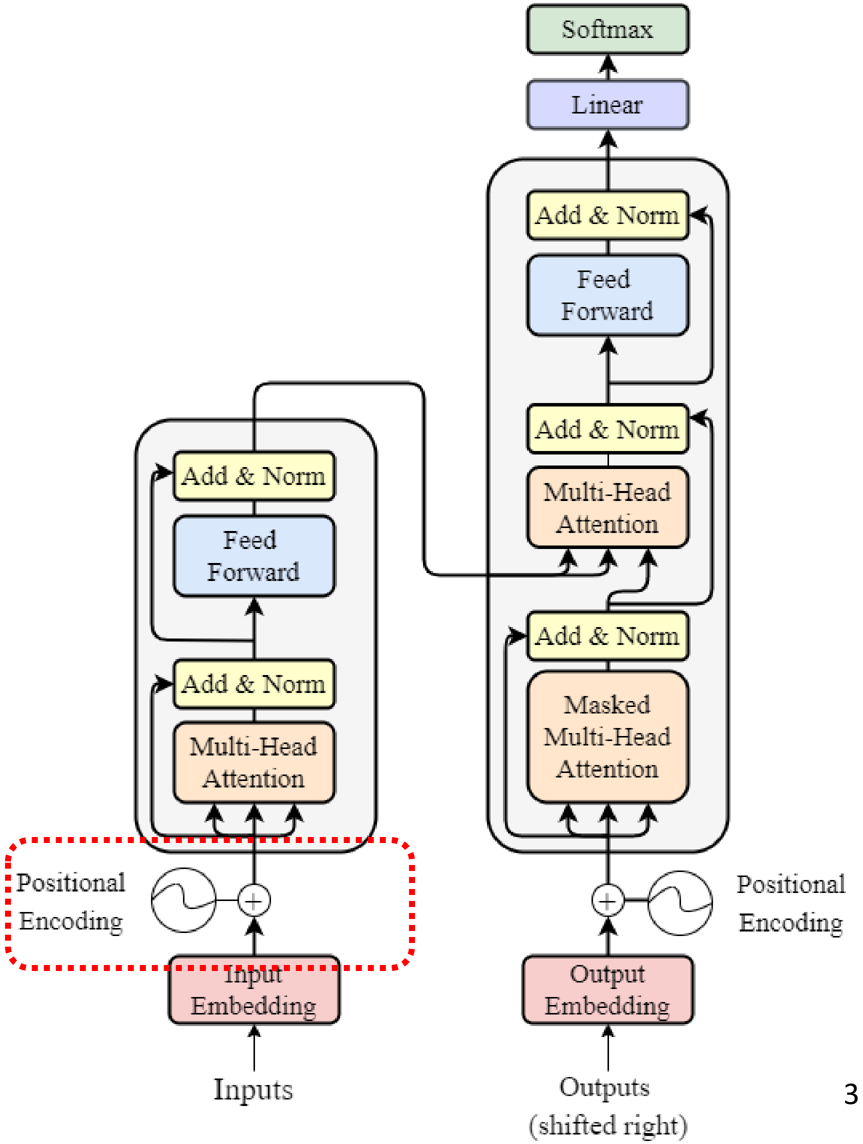


2. Position Embedding

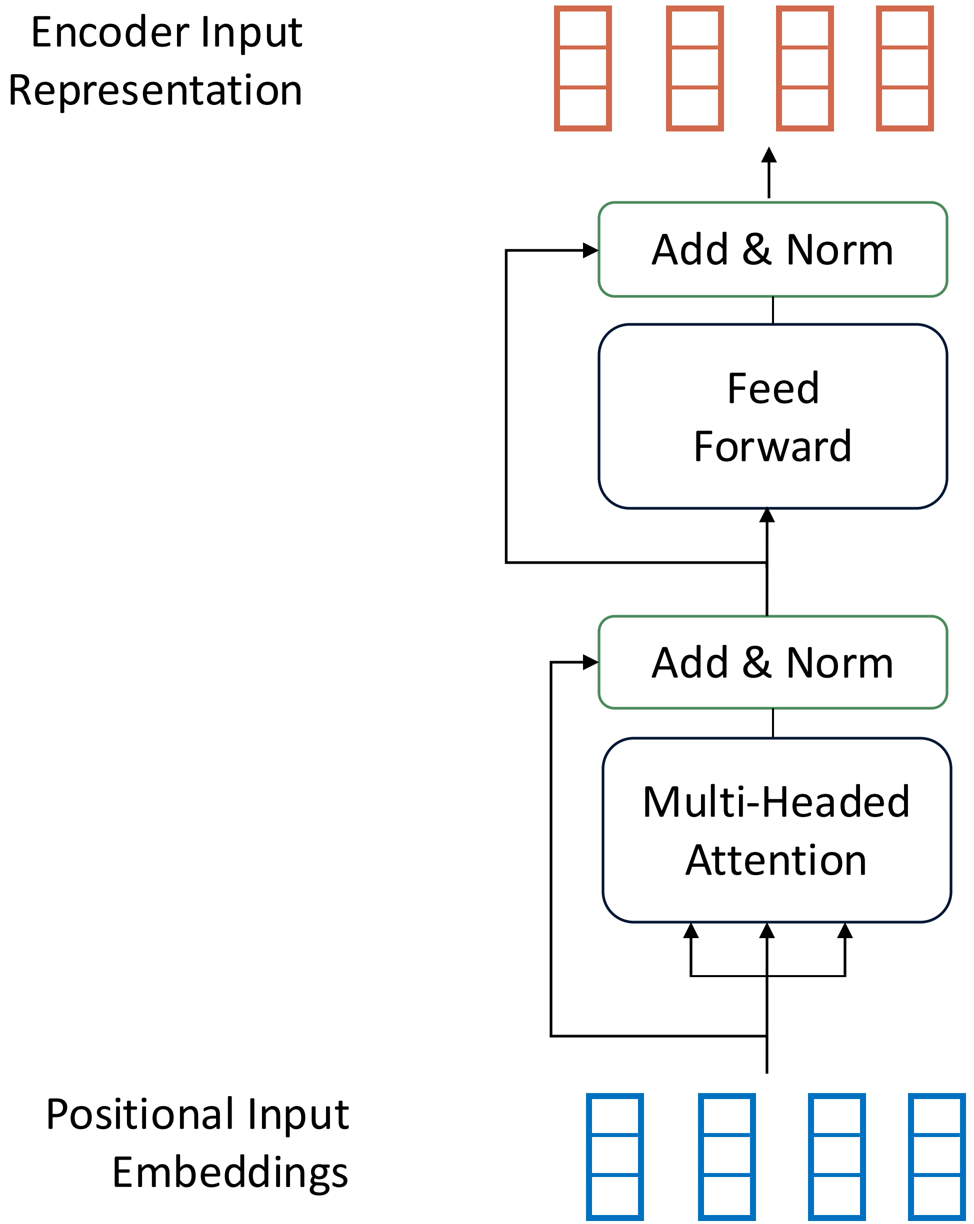


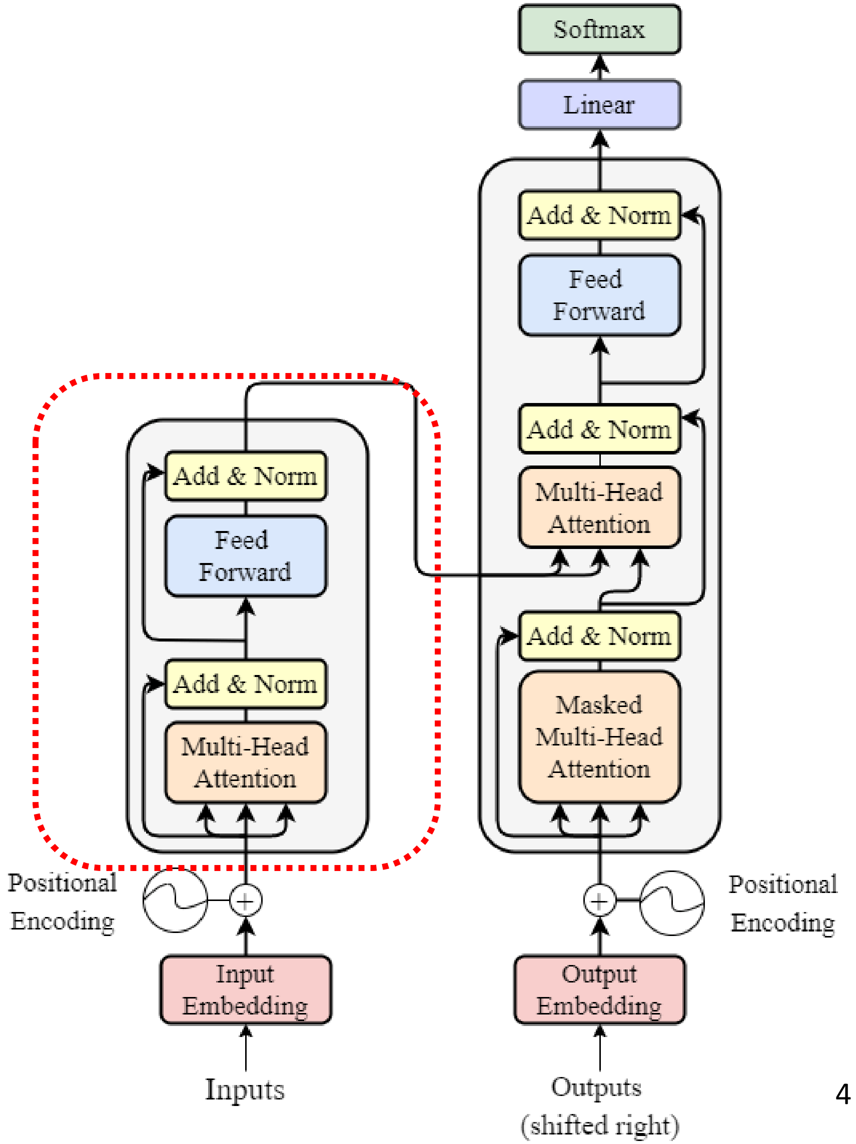
$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE(sin, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

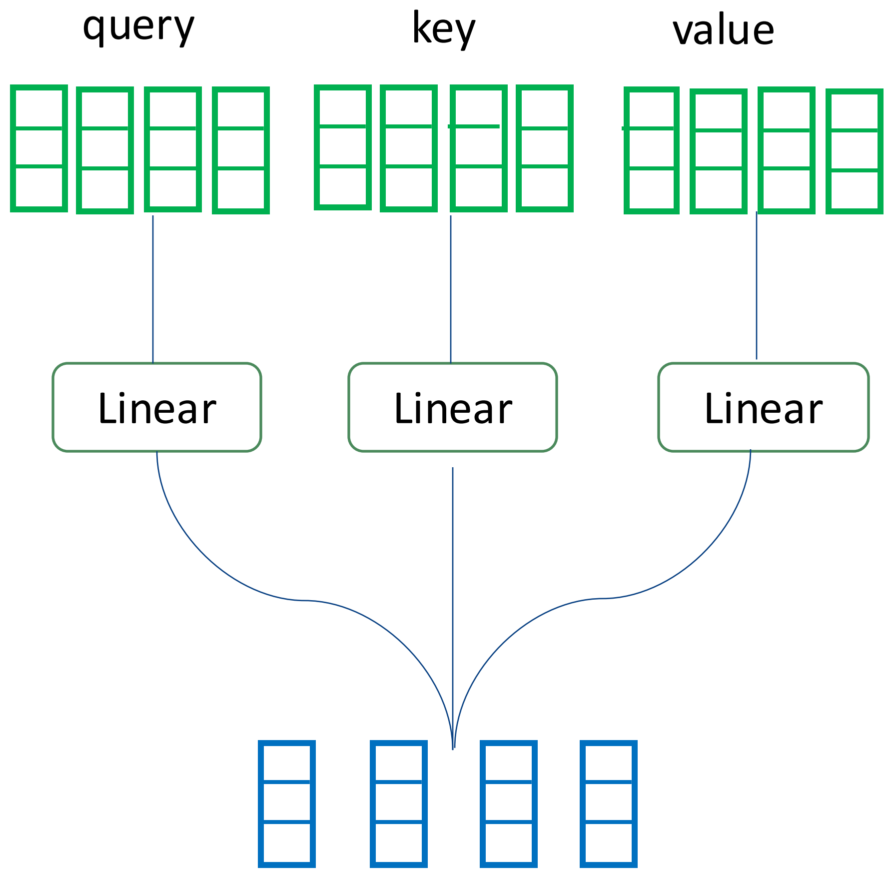


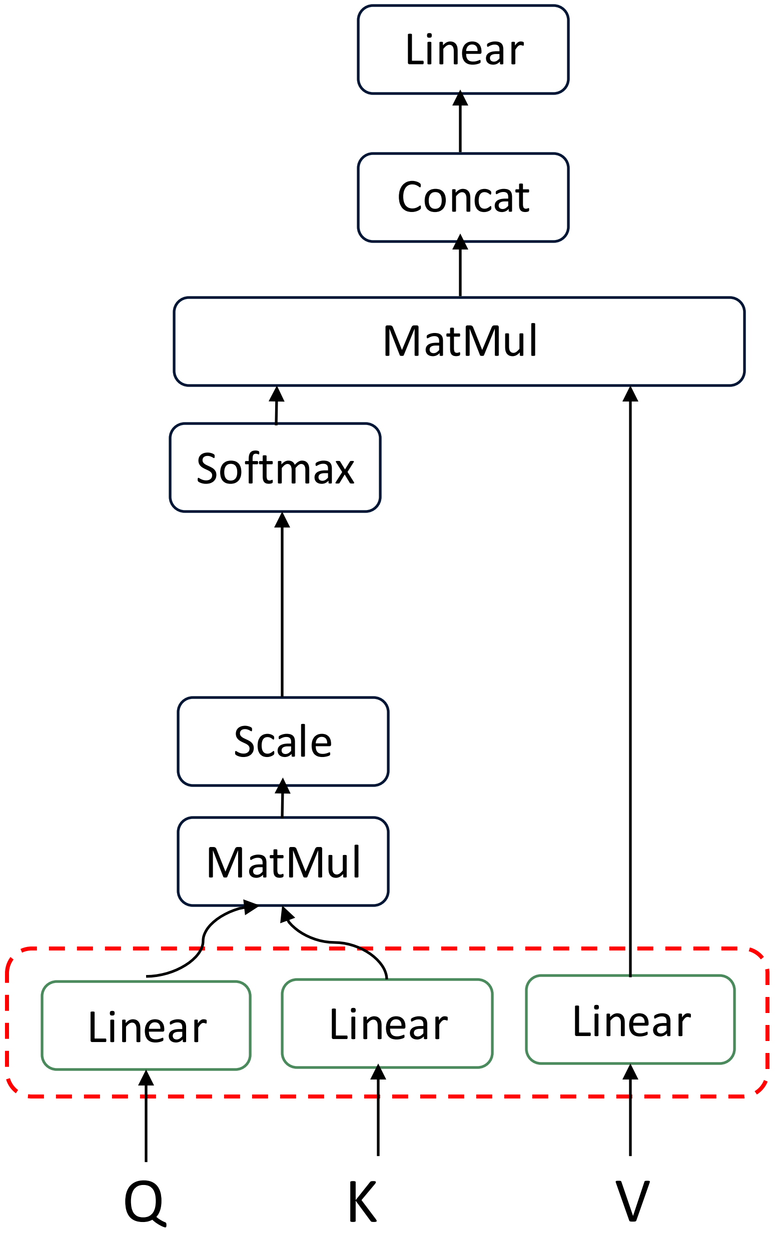
3-4. Encoder Layer



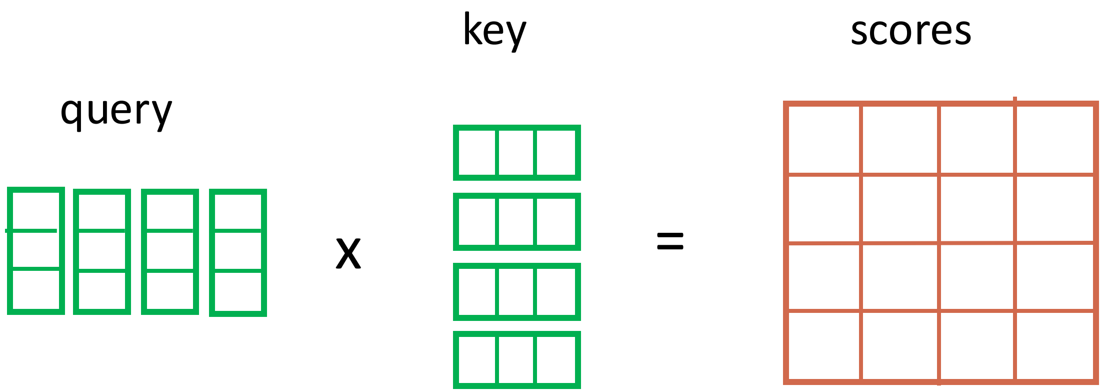


3. Multi-headed Attention

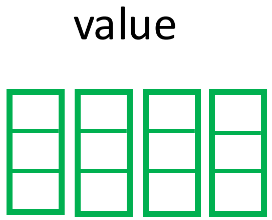


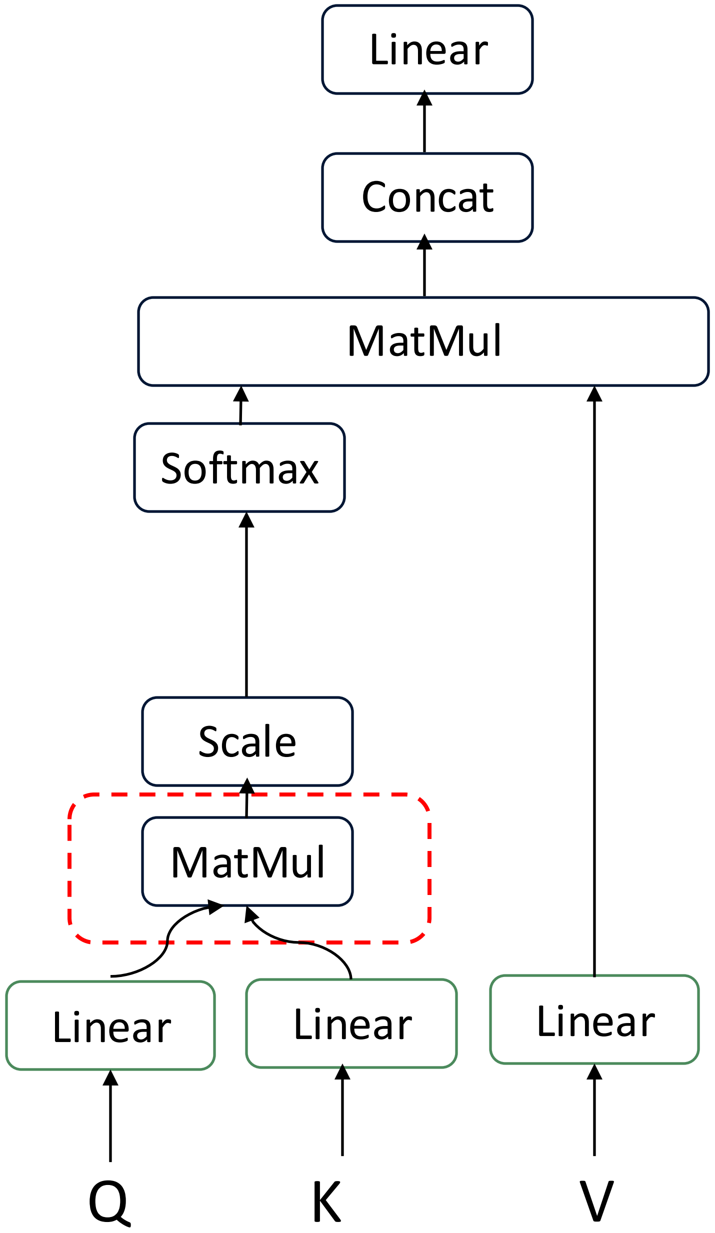


3. Multi-headed Attention
3.1 Self-Attention



	Hi	how	are	you
Hi	0.5	0.1	0.5	0.3
how	0.5	0.7	0.7	0.1
are	0.3	0.3	0.1	0.6
you	0.1	0.2	0.7	0.3





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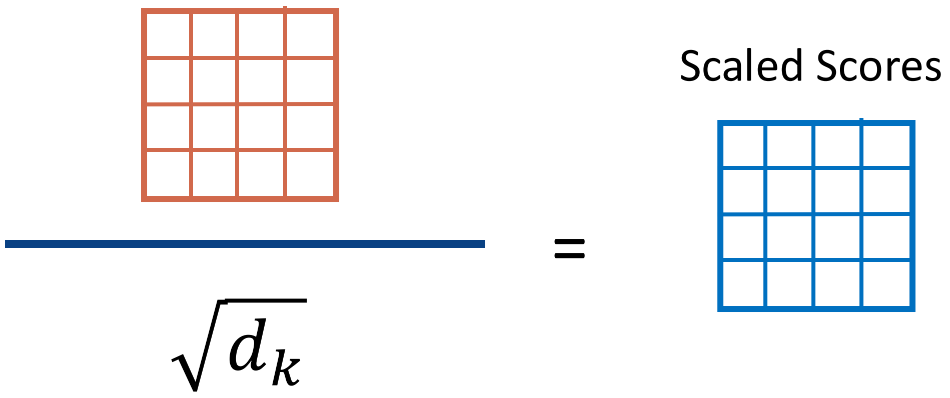
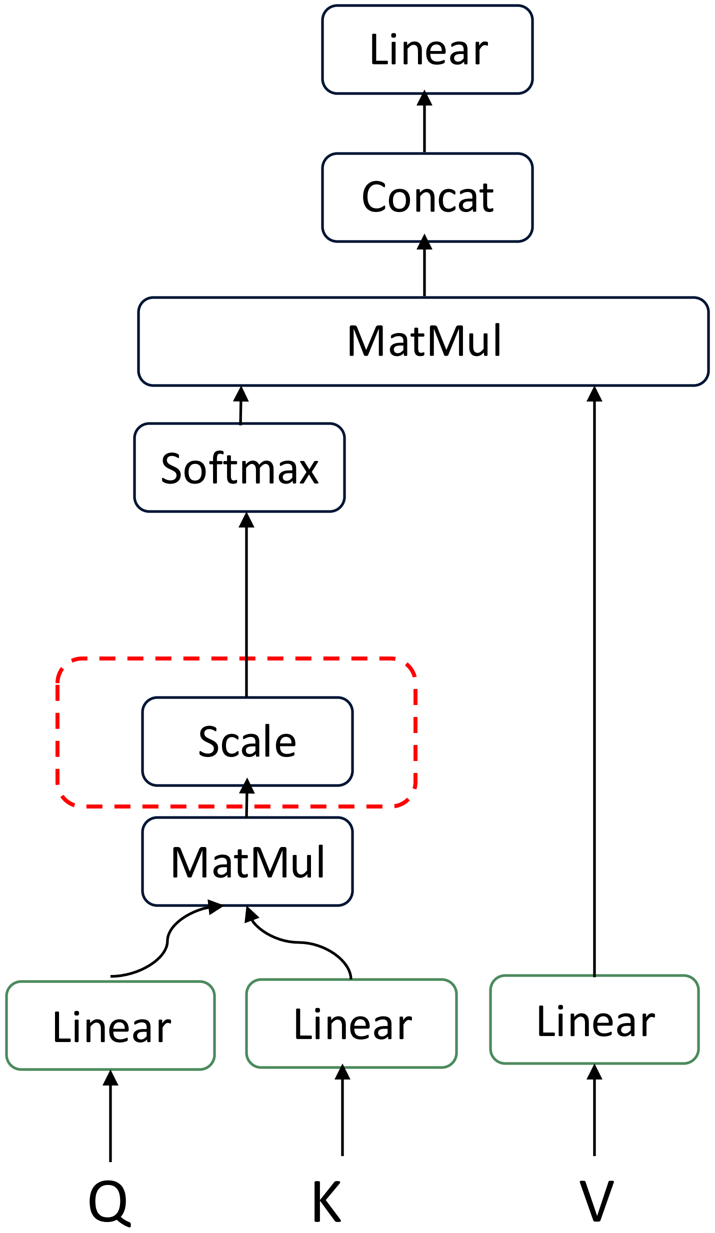
3.1 Self-Attention





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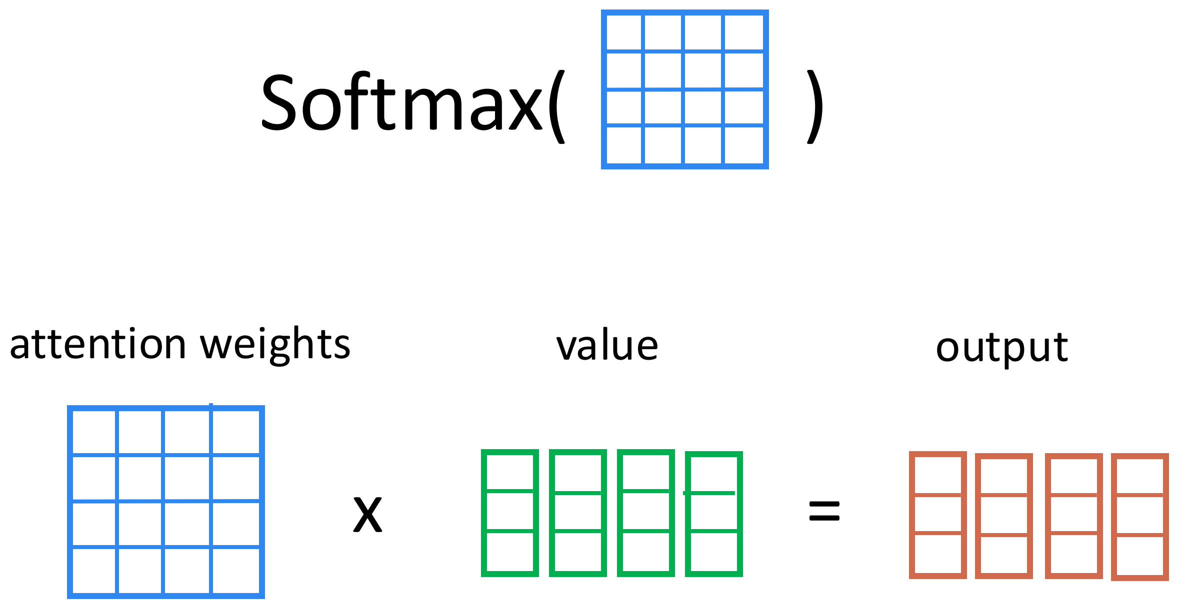
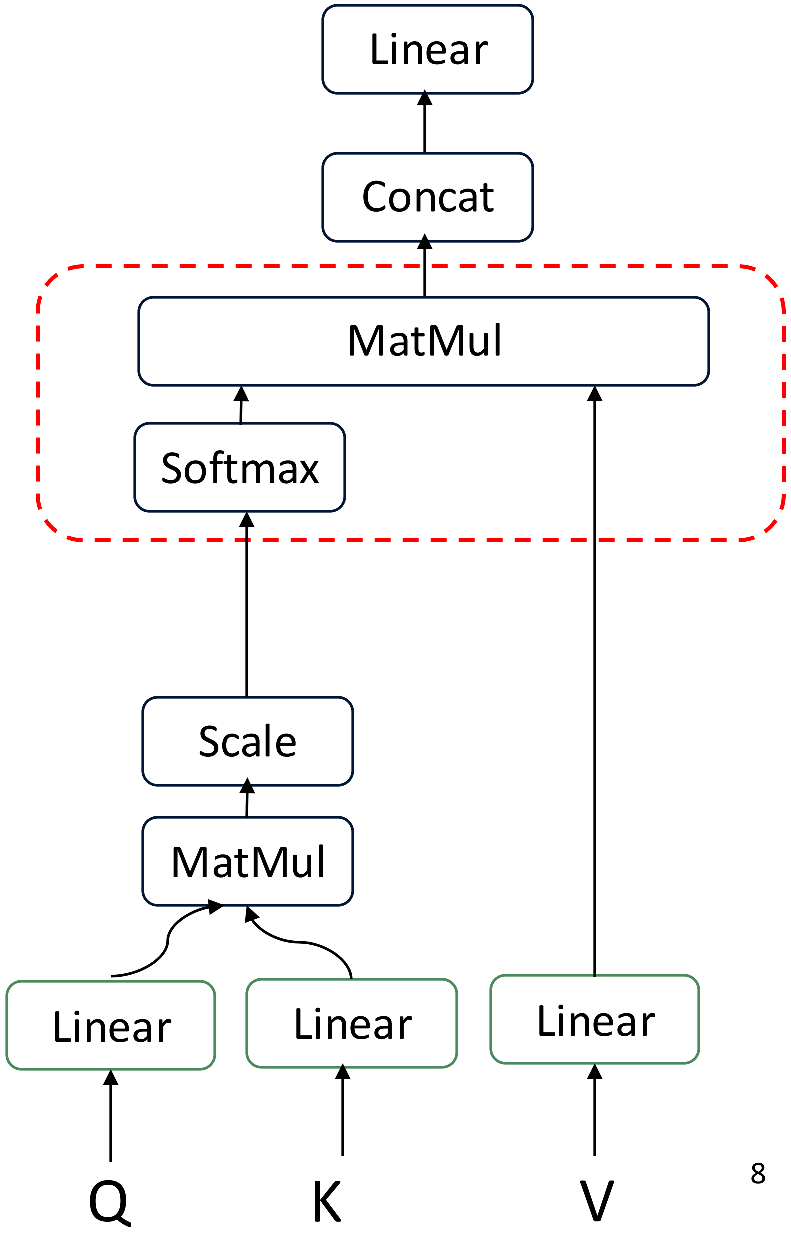
3.1 Self-Attention





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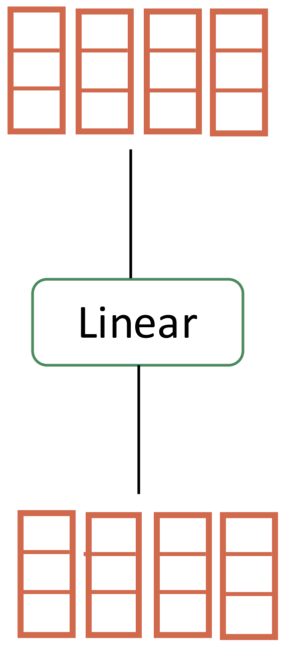
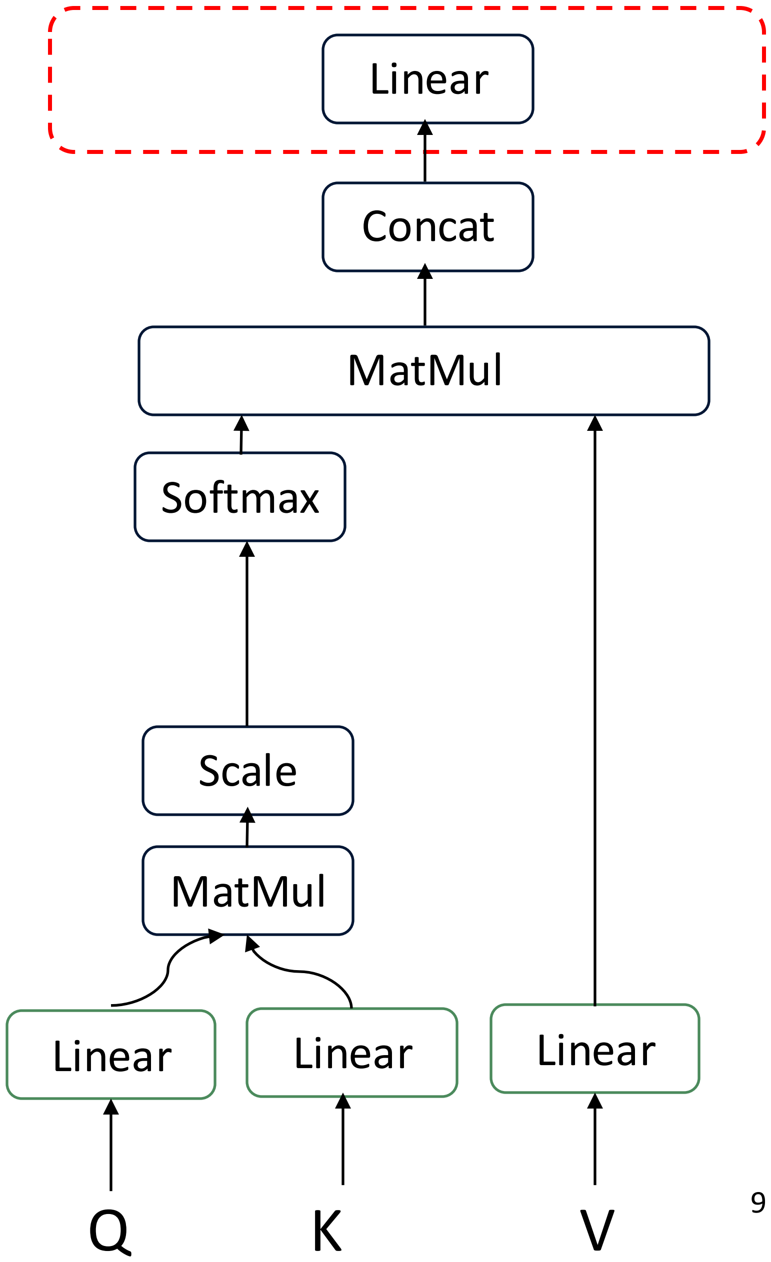
3.1 Self-Attention





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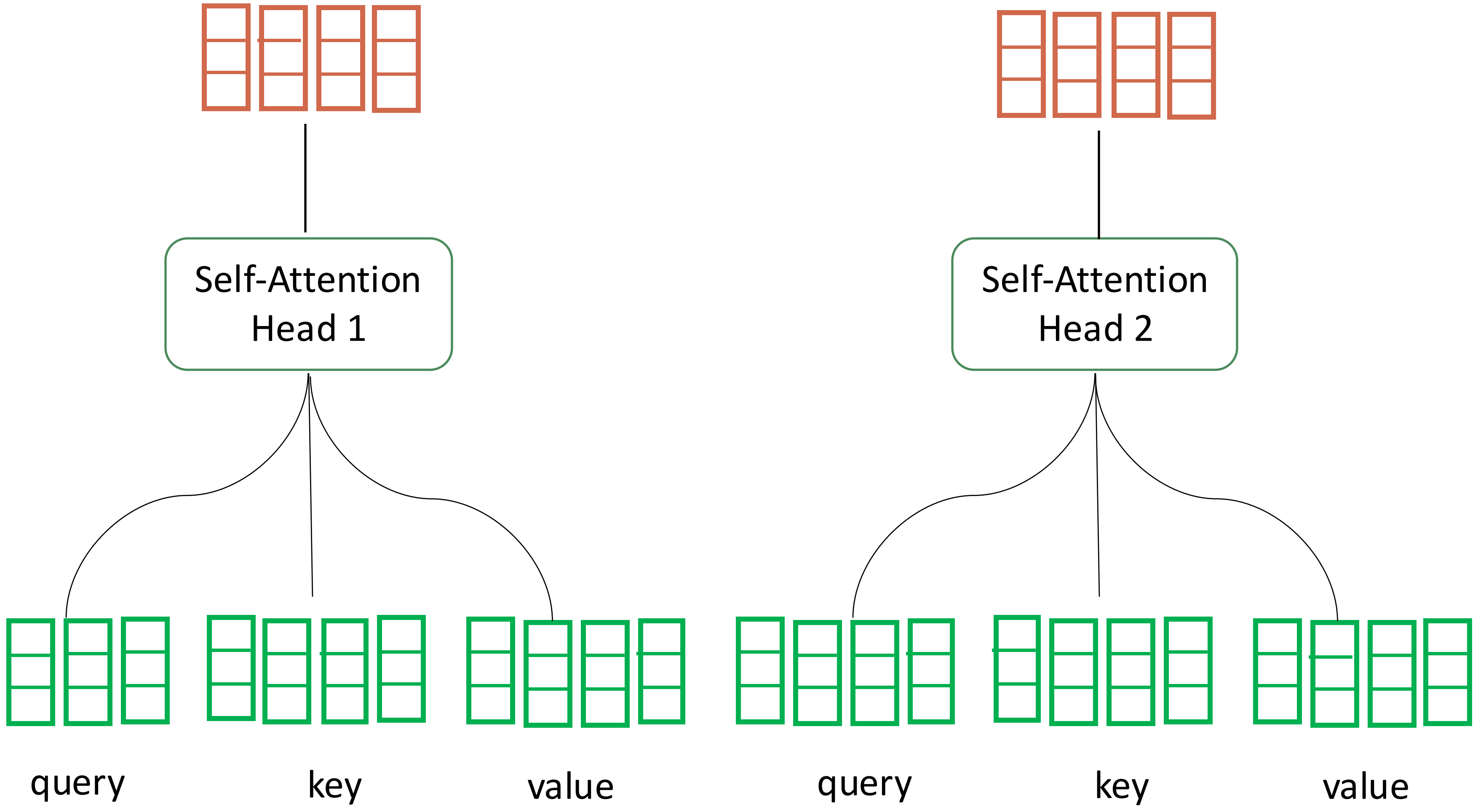
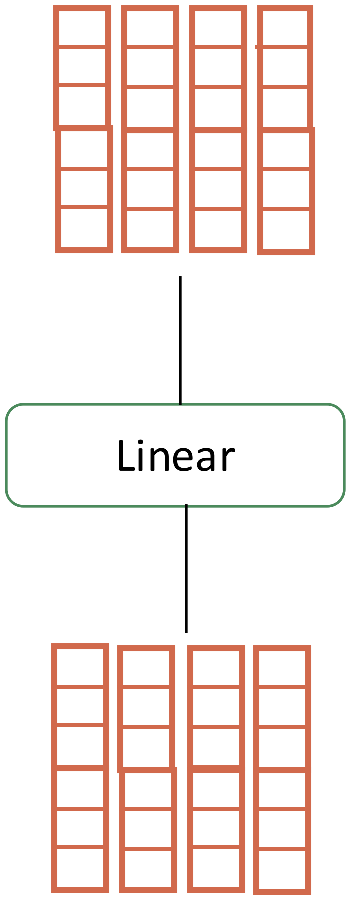
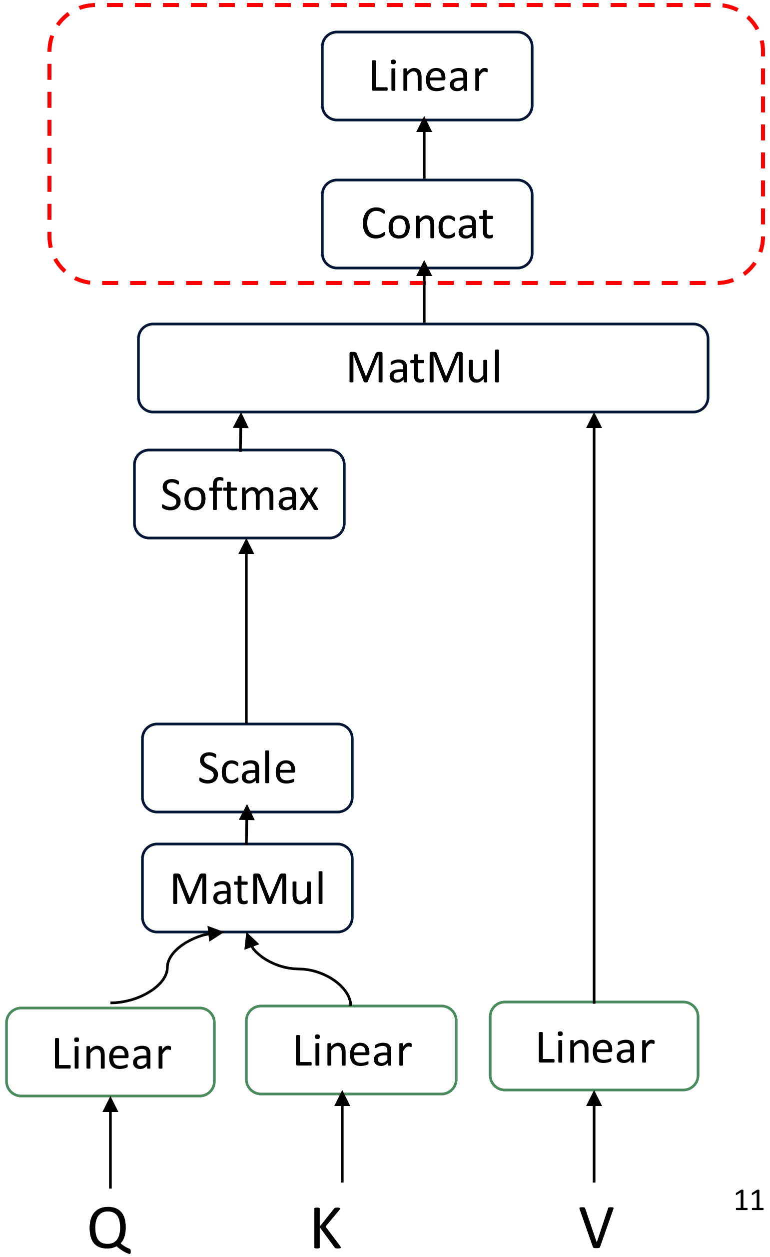


N = 2

3. Multi-headed Attention





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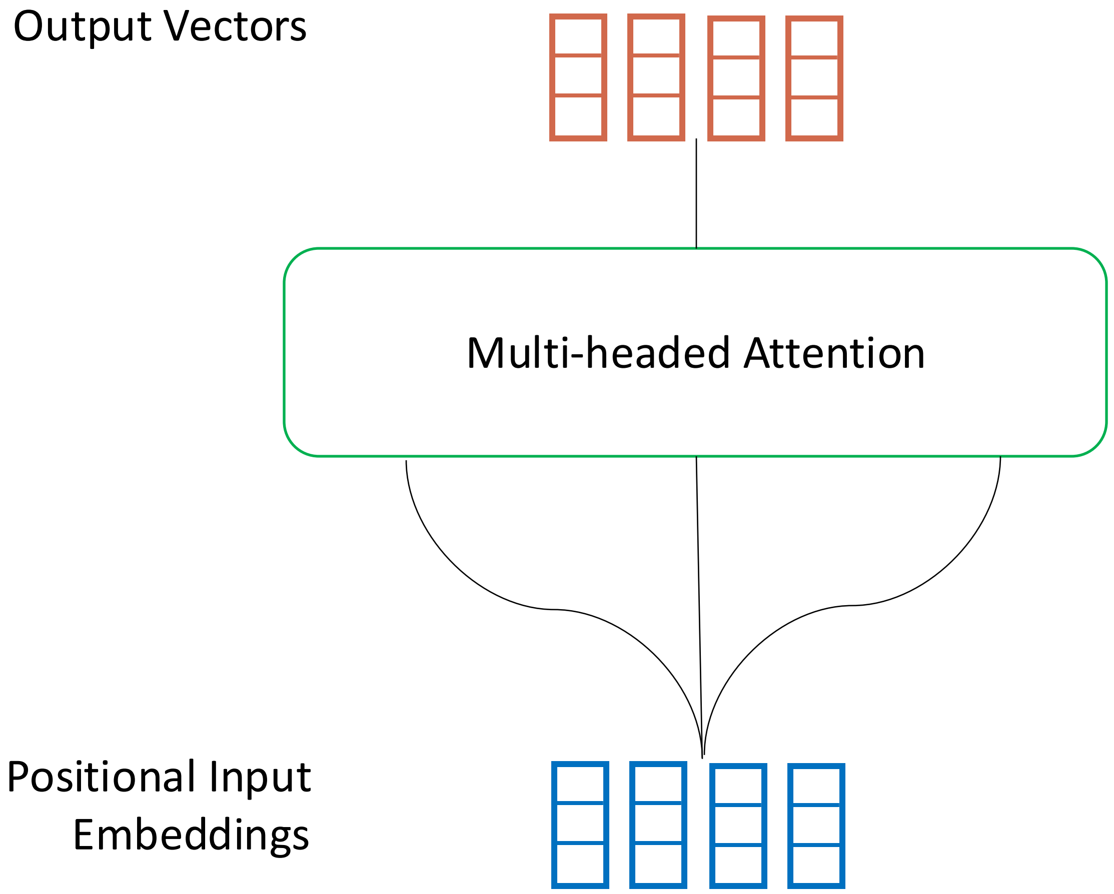
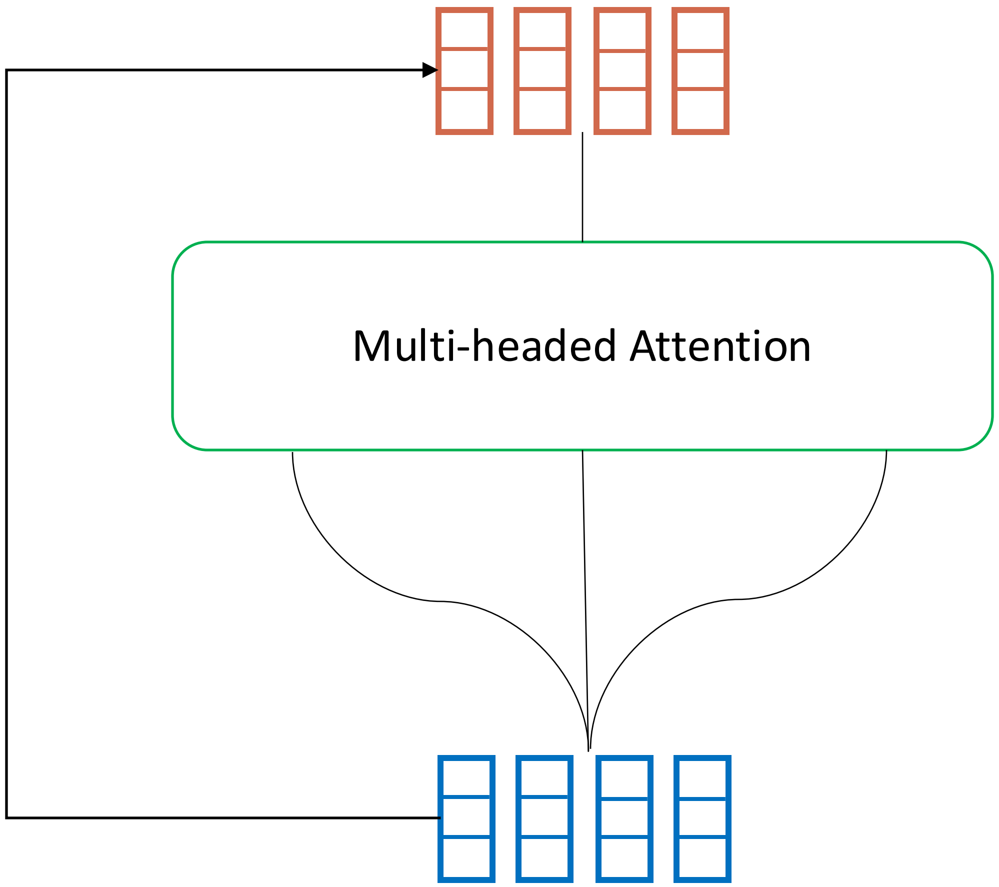
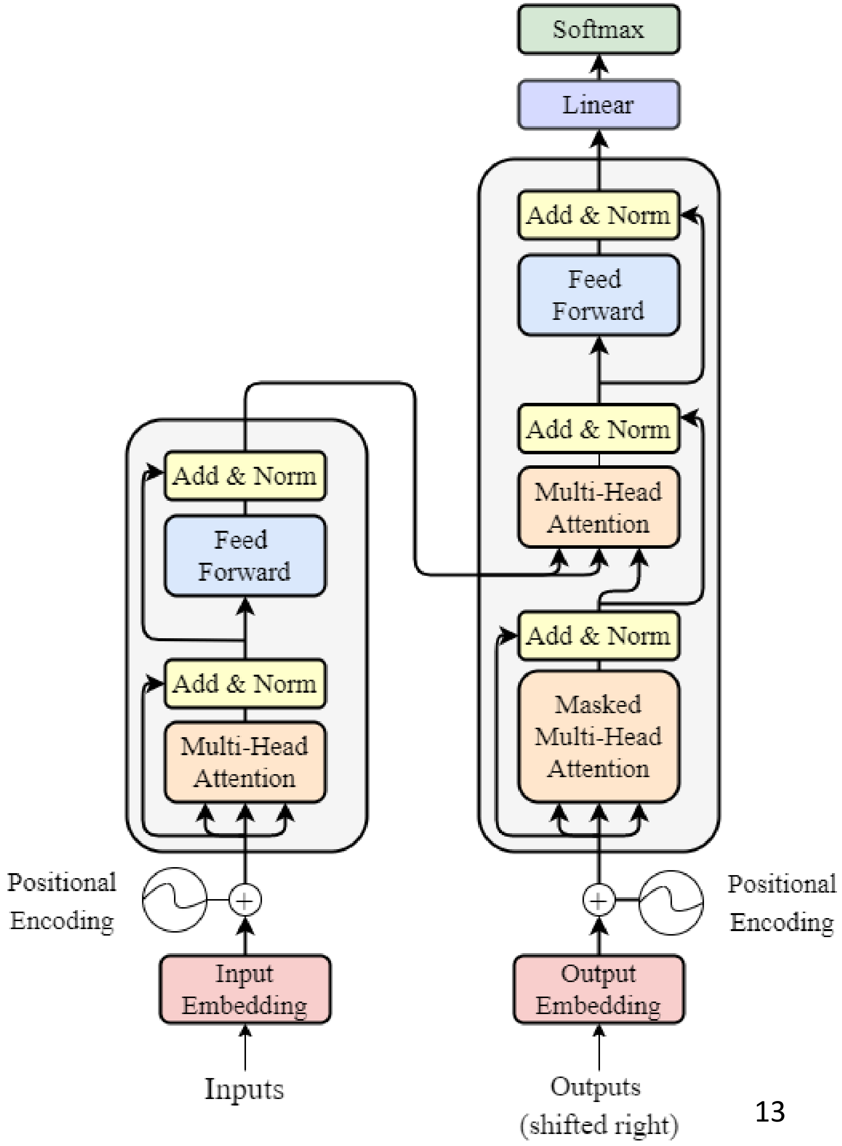


4. Residual Connection, Layer Normalization & Pointwise Feed Forward





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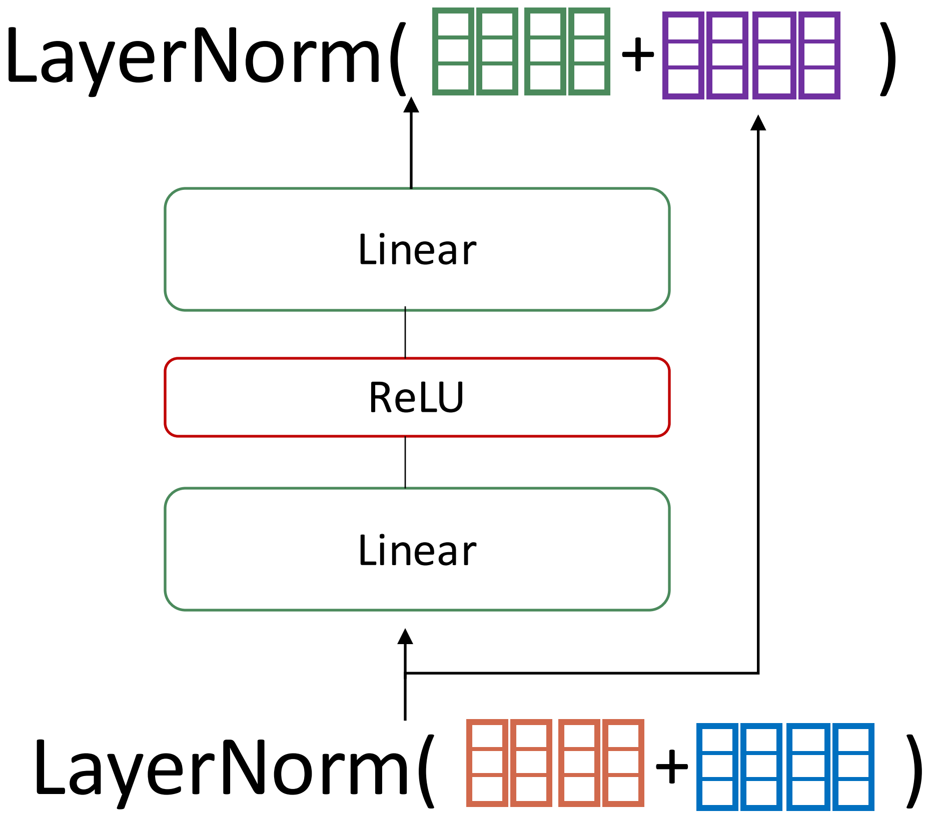
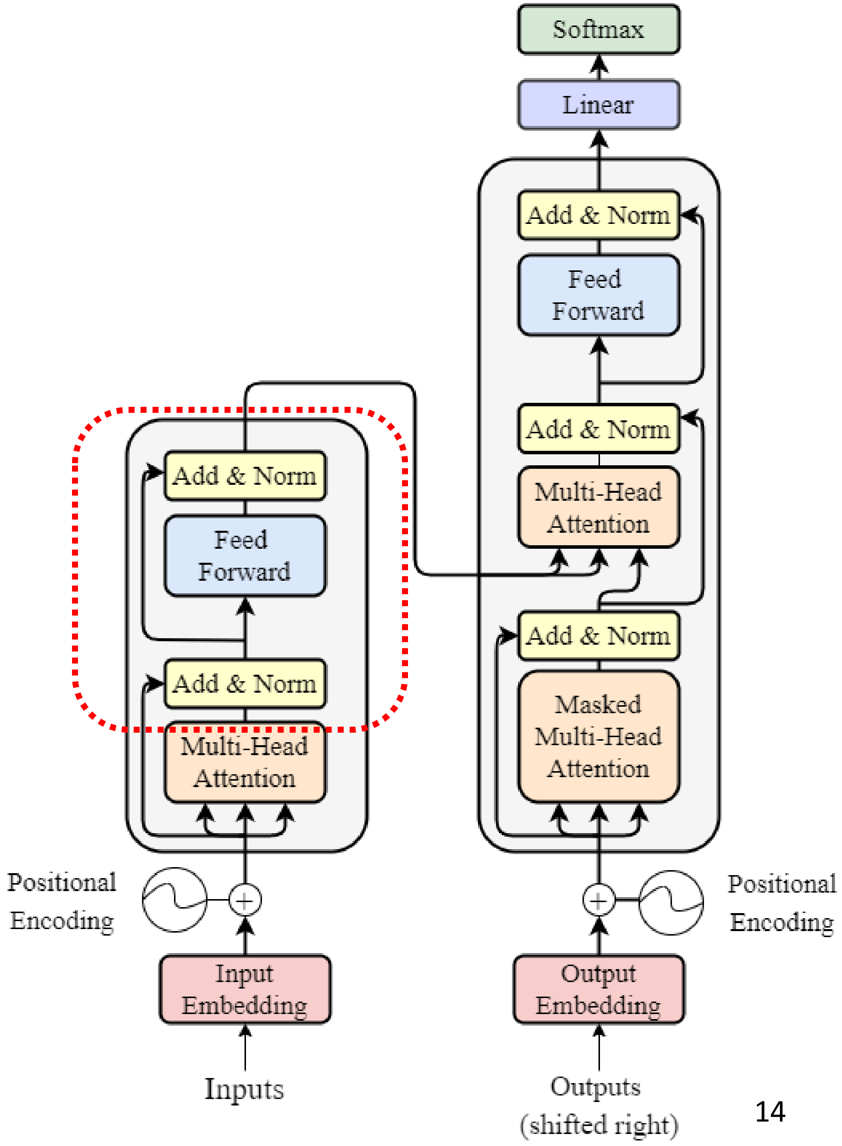
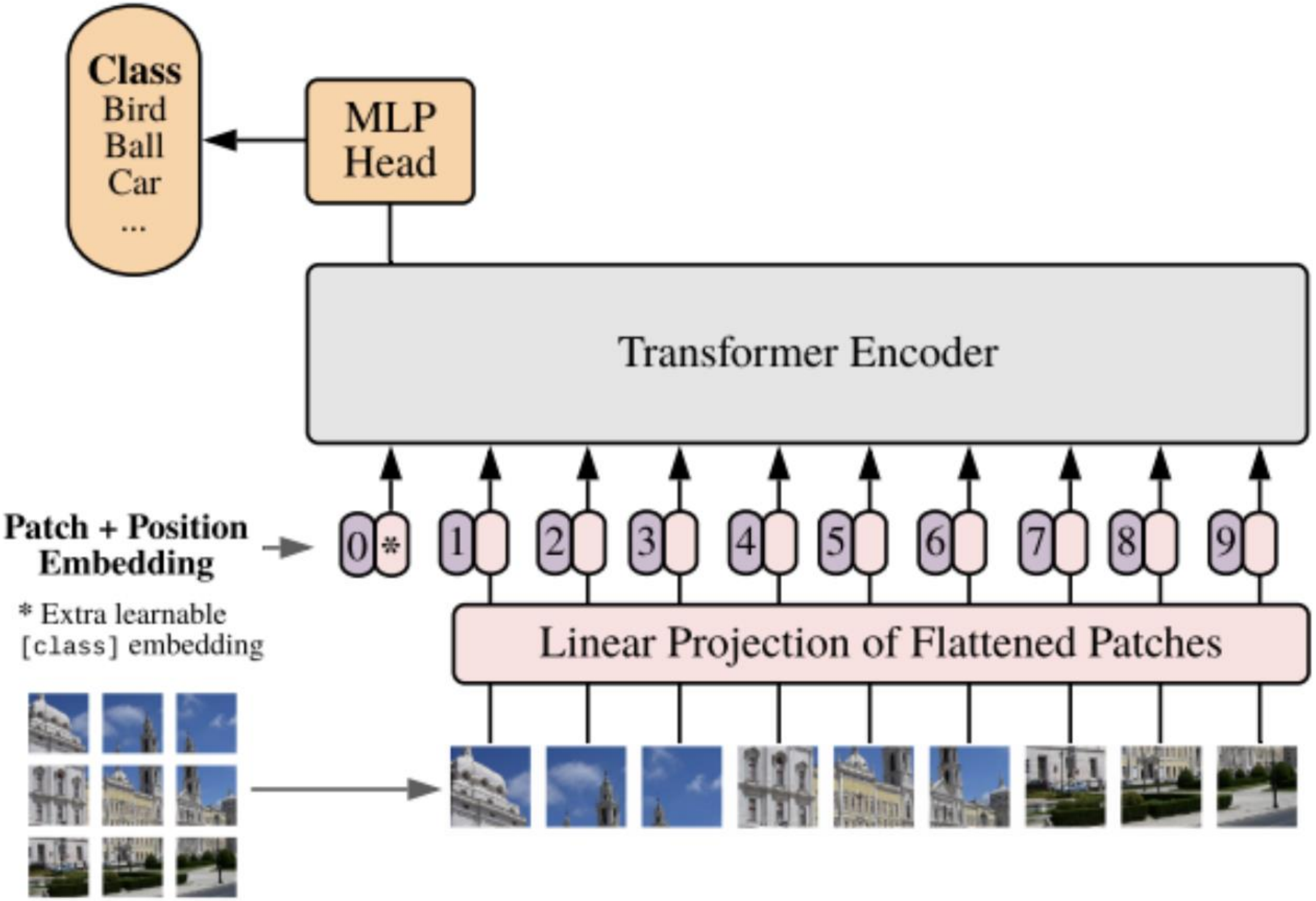
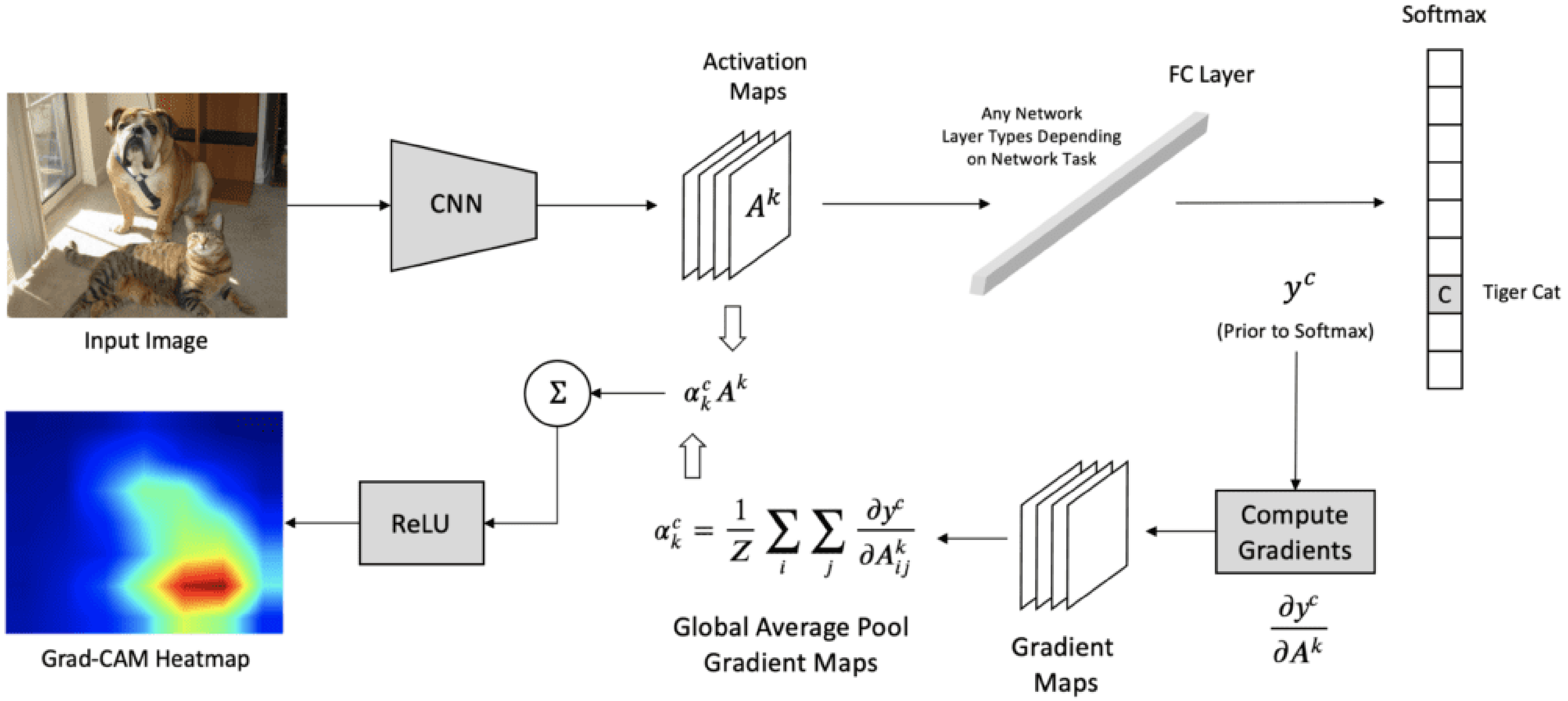




Vision Transformer (ViT)



Grad-CAM for CNNs



Applications of Grad-CAM in Medical Domain

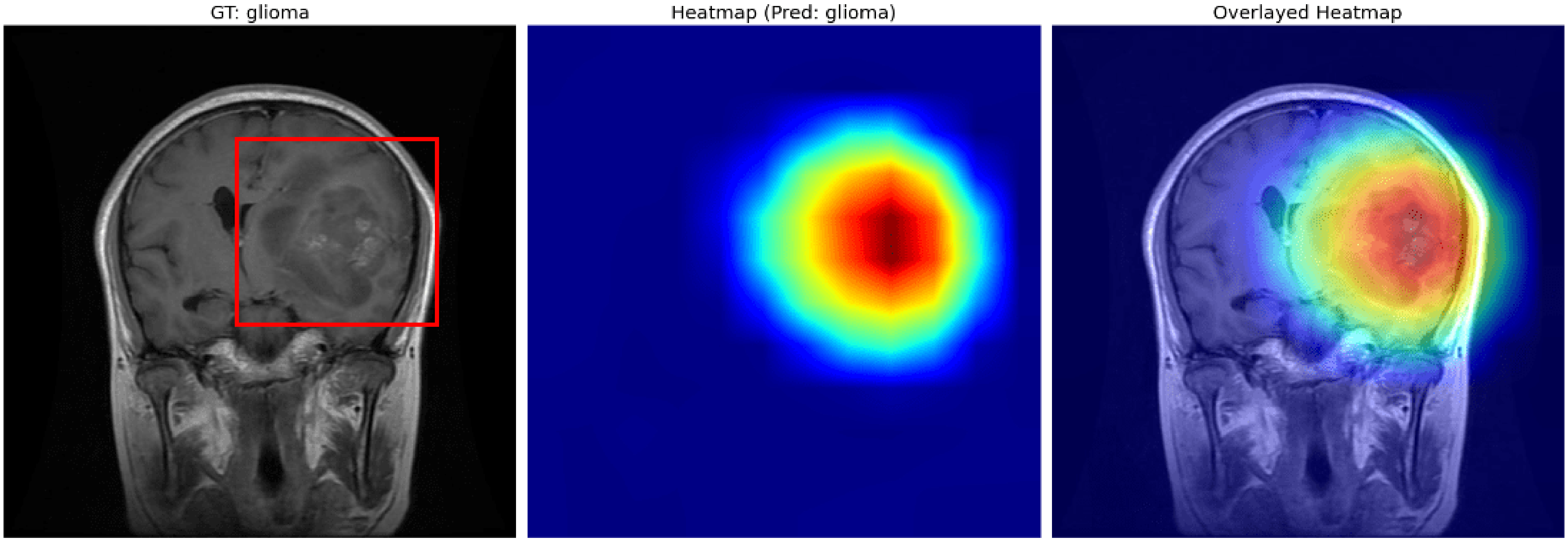
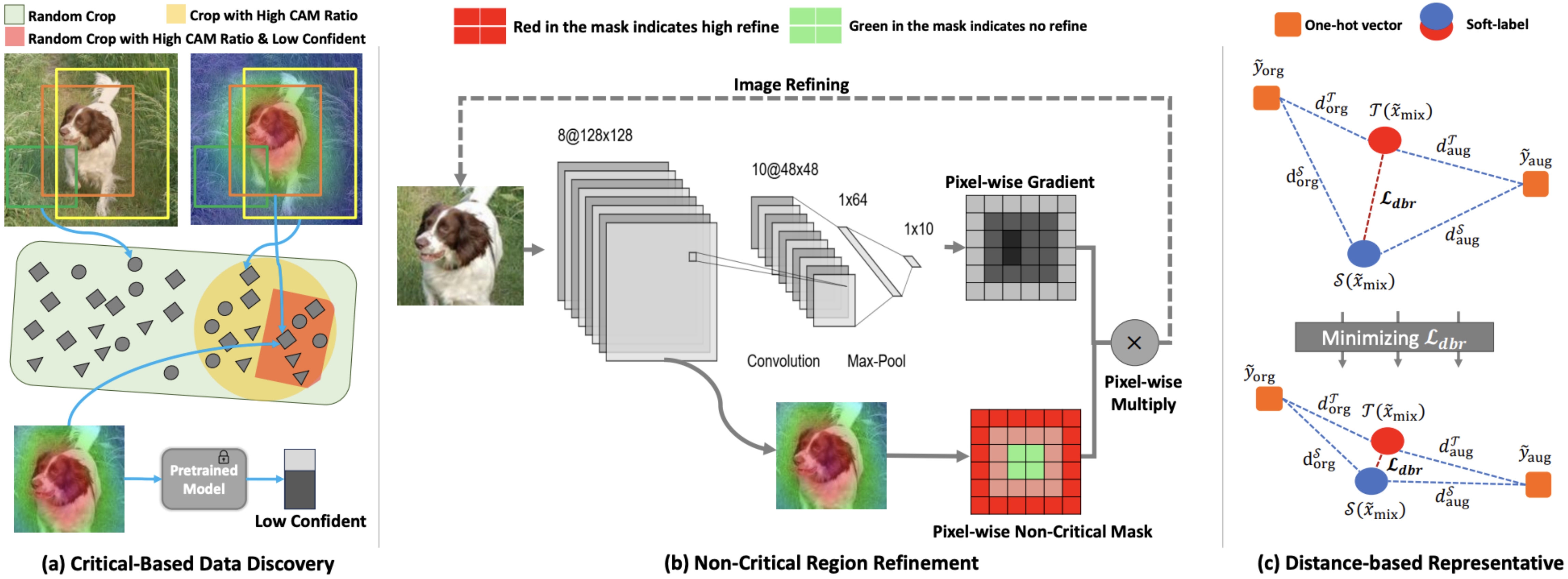


*Image from Google



Enhancing Dataset Distillation via Non-Critical Region Refinement. In *Proceedings of the Computer Vision and Pattern Recognition Conference* (CVPR 2025, A* conference)

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